

Database Schema – School Timetable System

1) Core Academic Context Tables

هذه الجداول تحدد السياق العام الذي يعمل داخله الجدول.

Table: schools

```
CREATE TABLE schools (  
    id BIGINT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,  
    name_ar VARCHAR(255) NOT NULL,  
    name_en VARCHAR(255),  
    code VARCHAR(100) UNIQUE NOT NULL,  
    timezone VARCHAR(100) NOT NULL DEFAULT 'Asia/Riyadh',  
    status VARCHAR(50) NOT NULL DEFAULT 'active',  
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,  
    updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP  
);
```

Table: academic_years

```
CREATE TABLE academic_years (  
    id BIGINT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,  
    school_id BIGINT NOT NULL REFERENCES schools(id) ON DELETE CASCADE,  
    name_ar VARCHAR(255) NOT NULL,  
    name_en VARCHAR(255),  
    start_date DATE NOT NULL,  
    end_date DATE NOT NULL,  
    is_active BOOLEAN NOT NULL DEFAULT FALSE,
```

```
status VARCHAR(50) NOT NULL DEFAULT 'active',  
  
created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP  
  
);
```

Table: terms

```
CREATE TABLE terms (  
  
    id BIGINT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,  
  
    school_id BIGINT NOT NULL REFERENCES schools(id) ON DELETE CASCADE,  
  
    academic_year_id BIGINT NOT NULL REFERENCES academic_years(id) ON DELETE  
CASCADE,  
  
    name_ar VARCHAR(255) NOT NULL,  
  
    name_en VARCHAR(255),  
  
    code VARCHAR(50),  
  
    start_date DATE NOT NULL,  
  
    end_date DATE NOT NULL,  
  
    is_active BOOLEAN NOT NULL DEFAULT FALSE,  
  
    status VARCHAR(50) NOT NULL DEFAULT 'active',  
  
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
    updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP  
  
);
```

2) Timetable Configuration Tables

هذه الجداول تمثل إعدادات تشغيل الجدول القادمة من صفحة إعدادات المدرسة.

Table: school_timetable_settings

```
CREATE TABLE school_timetable_settings (  
  
    id BIGINT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,  
  
    school_id BIGINT NOT NULL REFERENCES schools(id) ON DELETE CASCADE,  
  
    academic_year_id BIGINT NOT NULL REFERENCES academic_years(id) ON DELETE  
CASCADE,
```

term_id BIGINT NOT NULL REFERENCES terms(id) ON DELETE CASCADE,

week_start_day SMALLINT NOT NULL DEFAULT 1,

number_of_teaching_days SMALLINT NOT NULL DEFAULT 5,

periods_per_day SMALLINT NOT NULL DEFAULT 7,

default_period_duration_minutes SMALLINT NOT NULL DEFAULT 45,

school_day_start_time TIME NOT NULL,

school_day_end_time TIME NOT NULL,

max_daily_periods_per_teacher SMALLINT,

max_weekly_periods_per_teacher SMALLINT,

max_same_subject_per_day_per_class SMALLINT DEFAULT 2,

allow_teacher_gaps BOOLEAN NOT NULL DEFAULT TRUE,

allow_class_gaps BOOLEAN NOT NULL DEFAULT FALSE,

auto_balance_subjects BOOLEAN NOT NULL DEFAULT TRUE,

ai_generation_enabled BOOLEAN NOT NULL DEFAULT TRUE,

draft_generation_allowed BOOLEAN NOT NULL DEFAULT TRUE,

partial_regeneration_enabled BOOLEAN NOT NULL DEFAULT TRUE,

status VARCHAR(50) NOT NULL DEFAULT 'active',

created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,

updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,

UNIQUE (school_id, academic_year_id, term_id)

);

Table: school_weekdays

يمثل الأيام الدراسية المفعلة.

```
CREATE TABLE school_weekdays (  
    id BIGINT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,  
    school_id BIGINT NOT NULL REFERENCES schools(id) ON DELETE CASCADE,  
    academic_year_id BIGINT NOT NULL REFERENCES academic_years(id) ON DELETE CASCADE,  
    term_id BIGINT NOT NULL REFERENCES terms(id) ON DELETE CASCADE,  
  
    weekday_number SMALLINT NOT NULL,  
    weekday_name_ar VARCHAR(50) NOT NULL,  
    weekday_name_en VARCHAR(50),  
    is_active BOOLEAN NOT NULL DEFAULT TRUE,  
    display_order SMALLINT NOT NULL,  
  
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,  
    updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
    UNIQUE (school_id, academic_year_id, term_id, weekday_number)  
);
```

Table: school_time_blocks

هذا هو أهم جدول زمني أساسي.

يمثل كل البلوكات داخل اليوم: حصة، استراحة، صلاة، طاوور، بلوك مخصص.

```
CREATE TABLE school_time_blocks (  
    id BIGINT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,  
    school_id BIGINT NOT NULL REFERENCES schools(id) ON DELETE CASCADE,
```

```
academic_year_id BIGINT NOT NULL REFERENCES academic_years(id) ON DELETE
CASCADE,

term_id BIGINT NOT NULL REFERENCES terms(id) ON DELETE CASCADE,

block_type VARCHAR(50) NOT NULL,

title_ar VARCHAR(255) NOT NULL,

title_en VARCHAR(255),

sequence_no SMALLINT NOT NULL,

start_time TIME NOT NULL,

end_time TIME NOT NULL,

duration_minutes SMALLINT NOT NULL,

is_teaching_slot BOOLEAN NOT NULL DEFAULT TRUE,

is_active BOOLEAN NOT NULL DEFAULT TRUE,

affects_schedule_flow BOOLEAN NOT NULL DEFAULT TRUE,

color_token VARCHAR(50),

created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,

updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,

UNIQUE (school_id, academic_year_id, term_id, sequence_no)

);
```

block_type القيم المتوقعة في

- period
- break
- prayer
- assembly
- custom

3) Academic Structure Tables

Table: grades

```
CREATE TABLE grades (  
    id BIGINT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,  
    school_id BIGINT NOT NULL REFERENCES schools(id) ON DELETE CASCADE,  
    name_ar VARCHAR(255) NOT NULL,  
    name_en VARCHAR(255),  
    stage VARCHAR(100),  
    display_order SMALLINT NOT NULL,  
    is_active BOOLEAN NOT NULL DEFAULT TRUE,  
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,  
    updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP  
);
```

Table: classes

يمثل الشُّعب أو الفصول الفعلية.

```
CREATE TABLE classes (  
    id BIGINT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,  
    school_id BIGINT NOT NULL REFERENCES schools(id) ON DELETE CASCADE,  
    grade_id BIGINT NOT NULL REFERENCES grades(id) ON DELETE CASCADE,  
  
    name_ar VARCHAR(255) NOT NULL,  
    name_en VARCHAR(255),  
    section_code VARCHAR(50),  
    capacity INTEGER,  
    gender_type VARCHAR(50),  
    homeroom_teacher_id BIGINT NULL,
```

```
is_active BOOLEAN NOT NULL DEFAULT TRUE,  
  
created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP  
  
);
```

4) Subject Tables

Table: subjects

```
CREATE TABLE subjects (  
  
    id BIGINT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,  
  
    school_id BIGINT NOT NULL REFERENCES schools(id) ON DELETE CASCADE,  
  
  
  
    name_ar VARCHAR(255) NOT NULL,  
  
    name_en VARCHAR(255),  
  
    code VARCHAR(100),  
  
    category VARCHAR(100),  
  
    is_core BOOLEAN NOT NULL DEFAULT FALSE,  
  
    requires_lab BOOLEAN NOT NULL DEFAULT FALSE,  
  
    requires_special_room BOOLEAN NOT NULL DEFAULT FALSE,  
  
  
  
    is_active BOOLEAN NOT NULL DEFAULT TRUE,  
  
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
    updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP  
  
);
```

Table: grade_subjects

هذا الجدول يحدد المواد المطلوبة لكل صف وعدد حصصها الأسبوعية.

```
CREATE TABLE grade_subjects (  
  
    id BIGINT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,  
  
    school_id BIGINT NOT NULL REFERENCES schools(id) ON DELETE CASCADE,
```

```
grade_id BIGINT NOT NULL REFERENCES grades(id) ON DELETE CASCADE,  
subject_id BIGINT NOT NULL REFERENCES subjects(id) ON DELETE CASCADE,  
  
weekly_periods_required SMALLINT NOT NULL,  
priority_level SMALLINT DEFAULT 1,  
preferred_morning BOOLEAN NOT NULL DEFAULT FALSE,  
allow_consecutive BOOLEAN NOT NULL DEFAULT FALSE,  
max_periods_per_day SMALLINT DEFAULT 2,  
  
is_active BOOLEAN NOT NULL DEFAULT TRUE,  
created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,  
updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
UNIQUE (school_id, grade_id, subject_id)  
);
```

5) Teacher Tables

Table: teachers

```
CREATE TABLE teachers (  
    id BIGINT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,  
    school_id BIGINT NOT NULL REFERENCES schools(id) ON DELETE CASCADE,  
  
    user_id BIGINT,  
    employee_code VARCHAR(100),  
    full_name_ar VARCHAR(255) NOT NULL,  
    full_name_en VARCHAR(255),  
    status VARCHAR(50) NOT NULL DEFAULT 'active',  
    is_active BOOLEAN NOT NULL DEFAULT TRUE,
```

```
created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,  
updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP  
);
```

Table: teacher_subjects

هذا الجدول يربط المعلم بالمادة والصف الذي يمكنه تدريسه.

```
CREATE TABLE teacher_subjects (  
    id BIGINT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,  
    school_id BIGINT NOT NULL REFERENCES schools(id) ON DELETE CASCADE,  
    teacher_id BIGINT NOT NULL REFERENCES teachers(id) ON DELETE CASCADE,  
    subject_id BIGINT NOT NULL REFERENCES subjects(id) ON DELETE CASCADE,  
    grade_id BIGINT REFERENCES grades(id) ON DELETE CASCADE,  
  
    priority_rank SMALLINT DEFAULT 1,  
    is_primary_teacher BOOLEAN NOT NULL DEFAULT FALSE,  
    is_active BOOLEAN NOT NULL DEFAULT TRUE,  
  
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,  
    updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
    UNIQUE (school_id, teacher_id, subject_id, grade_id)  
);
```

Table: teacher_workload_rules

```
CREATE TABLE teacher_workload_rules (  
    id BIGINT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,  
    school_id BIGINT NOT NULL REFERENCES schools(id) ON DELETE CASCADE,  
    teacher_id BIGINT NOT NULL REFERENCES teachers(id) ON DELETE CASCADE,
```

```
max_periods_per_day SMALLINT,  
  
max_periods_per_week SMALLINT,  
  
max_consecutive_periods SMALLINT,  
  
min_free_periods_per_week SMALLINT,  
  
  
  
preferred_first_period BOOLEAN,  
  
preferred_last_period BOOLEAN,  
  
avoid_first_period BOOLEAN,  
  
avoid_last_period BOOLEAN,  
  
  
  
created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,  
updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
  
UNIQUE (school_id, teacher_id)  
);
```

6) Availability and Constraint Tables

Table: teacher_unavailability

```
CREATE TABLE teacher_unavailability (  
  
    id BIGINT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,  
  
    school_id BIGINT NOT NULL REFERENCES schools(id) ON DELETE CASCADE,  
  
    teacher_id BIGINT NOT NULL REFERENCES teachers(id) ON DELETE CASCADE,  
  
  
  
    academic_year_id BIGINT REFERENCES academic_years(id) ON DELETE CASCADE,  
  
    term_id BIGINT REFERENCES terms(id) ON DELETE CASCADE,  
  
  
  
    weekday_number SMALLINT NOT NULL,
```

time_block_id BIGINT NOT NULL REFERENCES school_time_blocks(id) ON DELETE CASCADE,
reason VARCHAR(255),

created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

);

Table: class_unavailability

CREATE TABLE class_unavailability (

id BIGINT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,

school_id BIGINT NOT NULL REFERENCES schools(id) ON DELETE CASCADE,

class_id BIGINT NOT NULL REFERENCES classes(id) ON DELETE CASCADE,

academic_year_id BIGINT REFERENCES academic_years(id) ON DELETE CASCADE,

term_id BIGINT REFERENCES terms(id) ON DELETE CASCADE,

weekday_number SMALLINT NOT NULL,

time_block_id BIGINT NOT NULL REFERENCES school_time_blocks(id) ON DELETE CASCADE,

reason VARCHAR(255),

created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

);

Table: subject_time_preferences

تفضيلات المادة على مستوى الوقت.

CREATE TABLE subject_time_preferences (

id BIGINT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,

school_id BIGINT NOT NULL REFERENCES schools(id) ON DELETE CASCADE,

subject_id BIGINT NOT NULL REFERENCES subjects(id) ON DELETE CASCADE,

```
academic_year_id BIGINT REFERENCES academic_years(id) ON DELETE CASCADE,  
term_id BIGINT REFERENCES terms(id) ON DELETE CASCADE,  
  
weekday_number SMALLINT,  
  
time_block_id BIGINT REFERENCES school_time_blocks(id) ON DELETE CASCADE,  
  
preference_type VARCHAR(50) NOT NULL,  
  
score_weight SMALLINT NOT NULL DEFAULT 10,  
  
created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP  
);
```

preference_type القيم المتوقعة في

- prefer
- avoid
- strong_prefer
- strong_avoid

Table: class_subject_constraints

قيود إضافية خاصة بمادة لفصل معين.

```
CREATE TABLE class_subject_constraints (  
  
id BIGINT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,  
  
school_id BIGINT NOT NULL REFERENCES schools(id) ON DELETE CASCADE,  
  
class_id BIGINT NOT NULL REFERENCES classes(id) ON DELETE CASCADE,  
  
subject_id BIGINT NOT NULL REFERENCES subjects(id) ON DELETE CASCADE,  
  
  
max_periods_per_day SMALLINT,  
  
allow_consecutive BOOLEAN,  
  
preferred_room_type VARCHAR(100),  
  
notes TEXT,
```

```
created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,  
updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
UNIQUE (school_id, class_id, subject_id)  
);
```

7) Room Tables — Optional but Recommended

Table: rooms

```
CREATE TABLE rooms (  
  
    id BIGINT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,  
    school_id BIGINT NOT NULL REFERENCES schools(id) ON DELETE CASCADE,  
  
    name VARCHAR(255) NOT NULL,  
    code VARCHAR(100),  
    room_type VARCHAR(100),  
    capacity INTEGER,  
    is_active BOOLEAN NOT NULL DEFAULT TRUE,  
  
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,  
    updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP  
);
```

Table: subject_room_requirements

```
CREATE TABLE subject_room_requirements (  
  
    id BIGINT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,  
    school_id BIGINT NOT NULL REFERENCES schools(id) ON DELETE CASCADE,  
    subject_id BIGINT NOT NULL REFERENCES subjects(id) ON DELETE CASCADE,
```

```
room_type VARCHAR(100) NOT NULL,  
  
is_mandatory BOOLEAN NOT NULL DEFAULT TRUE,  
  
created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP  
);
```

8) Timetable Versioning Tables

هذه الجداول تمثل النسخ الناتجة من التوليد.

Table: timetable_versions

```
CREATE TABLE timetable_versions (  
  
    id BIGINT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,  
  
    school_id BIGINT NOT NULL REFERENCES schools(id) ON DELETE CASCADE,  
  
    academic_year_id BIGINT NOT NULL REFERENCES academic_years(id) ON DELETE  
CASCADE,  
  
    term_id BIGINT NOT NULL REFERENCES terms(id) ON DELETE CASCADE,  
  
  
    version_name VARCHAR(255) NOT NULL,  
  
    generation_mode VARCHAR(50) NOT NULL,  
  
    status VARCHAR(50) NOT NULL DEFAULT 'draft',  
  
  
    quality_score NUMERIC(5,2),  
  
    conflicts_count INTEGER NOT NULL DEFAULT 0,  
  
    warnings_count INTEGER NOT NULL DEFAULT 0,  
  
  
    generated_by BIGINT,  
  
    generated_at TIMESTAMP,  
  
    published_at TIMESTAMP,
```

```
archived_at TIMESTAMP,  
  
notes TEXT,  
  
created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP  
);
```

status القيم المتوقعة في

- draft
- published
- archived
- failed

generation_mode القيم المتوقعة في

- full
- partial_class
- partial_teacher
- partial_day
- assist_mode

Table: timetable_entries

هذا هو الجدول الفعلي الناتج.

```
CREATE TABLE timetable_entries (  
  
    id BIGINT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,  
  
    school_id BIGINT NOT NULL REFERENCES schools(id) ON DELETE CASCADE,  
  
    academic_year_id BIGINT NOT NULL REFERENCES academic_years(id) ON DELETE  
CASCADE,  
  
    term_id BIGINT NOT NULL REFERENCES terms(id) ON DELETE CASCADE,  
  
    timetable_version_id BIGINT NOT NULL REFERENCES timetable_versions(id) ON DELETE  
CASCADE,  
  
    weekday_number SMALLINT NOT NULL,  
  
    time_block_id BIGINT NOT NULL REFERENCES school_time_blocks(id) ON DELETE CASCADE,
```

```

class_id BIGINT REFERENCES classes(id) ON DELETE CASCADE,
grade_id BIGINT REFERENCES grades(id) ON DELETE CASCADE,
subject_id BIGINT REFERENCES subjects(id) ON DELETE CASCADE,
teacher_id BIGINT REFERENCES teachers(id) ON DELETE CASCADE,
room_id BIGINT REFERENCES rooms(id) ON DELETE SET NULL,

entry_type VARCHAR(50) NOT NULL DEFAULT 'lecture',
entry_status VARCHAR(50) NOT NULL DEFAULT 'scheduled',
is_ai_generated BOOLEAN NOT NULL DEFAULT TRUE,
source VARCHAR(100) DEFAULT 'ai_engine',

notes TEXT,

created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
);

```

entry_type القيم المتوقعة في

- lecture
- break
- prayer
- assembly
- custom

entry_status القيم المتوقعة في

- scheduled
- modified
- locked
- cancelled

9) Timetable Diagnostics and Logs

Table: timetable_generation_runs

يمثل كل عملية تشغيل للمحرك.

```
CREATE TABLE timetable_generation_runs (  
    id BIGINT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,  
    school_id BIGINT NOT NULL REFERENCES schools(id) ON DELETE CASCADE,  
    academic_year_id BIGINT NOT NULL REFERENCES academic_years(id) ON DELETE  
CASCADE,  
    term_id BIGINT NOT NULL REFERENCES terms(id) ON DELETE CASCADE,  
  
    timetable_version_id BIGINT REFERENCES timetable_versions(id) ON DELETE SET NULL,  
  
    trigger_type VARCHAR(50) NOT NULL,  
    engine_version VARCHAR(100),  
    status VARCHAR(50) NOT NULL DEFAULT 'running',  
  
    total_units INTEGER NOT NULL DEFAULT 0,  
    assigned_units INTEGER NOT NULL DEFAULT 0,  
    unassigned_units INTEGER NOT NULL DEFAULT 0,  
  
    hard_conflicts INTEGER NOT NULL DEFAULT 0,  
    soft_conflicts INTEGER NOT NULL DEFAULT 0,  
  
    initiated_by BIGINT,  
    started_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,  
    finished_at TIMESTAMP,  
  
    metadata_json JSONB,  
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,  
    updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP  
);
```

Table: timetable_generation_issues

```
CREATE TABLE timetable_generation_issues (  
  
    id BIGINT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,  
  
    school_id BIGINT NOT NULL REFERENCES schools(id) ON DELETE CASCADE,  
  
    generation_run_id BIGINT NOT NULL REFERENCES timetable_generation_runs(id) ON DELETE CASCADE,  
  
  
  
    issue_type VARCHAR(100) NOT NULL,  
  
    severity VARCHAR(50) NOT NULL,  
  
    related_entity_type VARCHAR(100),  
  
    related_entity_id BIGINT,  
  
  
  
    message TEXT NOT NULL,  
  
    details_json JSONB,  
  
  
  
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP  
);
```

أمثلة issue_type

- missing_teacher_mapping
- insufficient_time_slots
- teacher_overload
- unassigned_unit
- room_conflict
- invalid_school_settings

Table: timetable_audit_logs

```
CREATE TABLE timetable_audit_logs (  
  
    id BIGINT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,  
  
    school_id BIGINT NOT NULL REFERENCES schools(id) ON DELETE CASCADE,  
  
    timetable_version_id BIGINT REFERENCES timetable_versions(id) ON DELETE CASCADE,
```

```

action_type VARCHAR(100) NOT NULL,

action_by BIGINT,

target_type VARCHAR(100),

target_id BIGINT,


old_value_json JSONB,

new_value_json JSONB,


created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
);

```

10) AI Support Tables — Recommended

Table: timetable_generation_preferences

في التقييم AI / Heuristic Engine هذا الجدول يخزن الأوزان التي يستخدمها الـ

```

CREATE TABLE timetable_generation_preferences (

    id BIGINT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,

    school_id BIGINT NOT NULL REFERENCES schools(id) ON DELETE CASCADE,

    academic_year_id BIGINT REFERENCES academic_years(id) ON DELETE CASCADE,

    term_id BIGINT REFERENCES terms(id) ON DELETE CASCADE,


    prefer_core_subjects_early_weight SMALLINT DEFAULT 15,

    avoid_teacher_gaps_weight SMALLINT DEFAULT 10,

    avoid_class_gaps_weight SMALLINT DEFAULT 10,

    balance_teacher_load_weight SMALLINT DEFAULT 10,

    balance_class_load_weight SMALLINT DEFAULT 10,

    avoid_consecutive_same_subject_weight SMALLINT DEFAULT 8,

    respect_teacher_preference_weight SMALLINT DEFAULT 7,

```

avoid_late_heavy_subjects_weight SMALLINT DEFAULT 6,

created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,

updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,

UNIQUE (school_id, academic_year_id, term_id)

);

11) Suggested Indexes

مهمة جدًا للأداء indexes هذه الـ

CREATE INDEX idx_terms_school_year ON terms(school_id, academic_year_id);

CREATE INDEX idx_time_blocks_school_term ON school_time_blocks(school_id, academic_year_id, term_id);

CREATE INDEX idx_weekdays_school_term ON school_weekdays(school_id, academic_year_id, term_id);

CREATE INDEX idx_classes_school_grade ON classes(school_id, grade_id);

CREATE INDEX idx_grade_subjects_school_grade ON grade_subjects(school_id, grade_id);

CREATE INDEX idx_teacher_subjects_school_teacher ON teacher_subjects(school_id, teacher_id);

CREATE INDEX idx_teacher_subjects_subject_grade ON teacher_subjects(subject_id, grade_id);

CREATE INDEX idx_teacher_unavailability_lookup

ON teacher_unavailability(school_id, teacher_id, weekday_number, time_block_id);

CREATE INDEX idx_class_unavailability_lookup

ON class_unavailability(school_id, class_id, weekday_number, time_block_id);

CREATE INDEX idx_timetable_versions_lookup

```
ON timetable_versions(school_id, academic_year_id, term_id, status);
```

```
CREATE INDEX idx_timetable_entries_version
```

```
ON timetable_entries(timetable_version_id);
```

```
CREATE INDEX idx_timetable_entries_teacher_slot
```

```
ON timetable_entries(timetable_version_id, teacher_id, weekday_number, time_block_id);
```

```
CREATE INDEX idx_timetable_entries_class_slot
```

```
ON timetable_entries(timetable_version_id, class_id, weekday_number, time_block_id);
```

```
CREATE INDEX idx_timetable_entries_subject_slot
```

```
ON timetable_entries(timetable_version_id, subject_id, weekday_number, time_block_id);
```

```
CREATE INDEX idx_generation_runs_version
```

```
ON timetable_generation_runs(timetable_version_id);
```

12) Suggested Integrity Rules

يفضل أيضًا إضافة قواعد منطقية على مستوى قاعدة البيانات:

منع تعارض المعلم داخل نفس النسخة

```
CREATE UNIQUE INDEX uq_teacher_one_slot_per_version
```

```
ON timetable_entries (timetable_version_id, teacher_id, weekday_number, time_block_id)
```

```
WHERE teacher_id IS NOT NULL AND entry_status <> 'cancelled';
```

منع تعارض الفصل داخل نفس النسخة

```
CREATE UNIQUE INDEX uq_class_one_slot_per_version
```

```
ON timetable_entries (timetable_version_id, class_id, weekday_number, time_block_id)
```

```
WHERE class_id IS NOT NULL AND entry_status <> 'cancelled';
```

منع تكرار نفس البلوك أكثر من مرة للفصل في نفس النسخة

```
CREATE UNIQUE INDEX uq_class_block_once_per_version  
ON timetable_entries (timetable_version_id, class_id, weekday_number, time_block_id, entry_type)  
WHERE class_id IS NOT NULL AND entry_status <> 'cancelled';
```

13) العلاقات الأساسية بين الجداول

العلاقات الرئيسية كالتالي:

- يمتلك school:
 - academic_years
 - terms
 - school_timetable_settings
 - school_weekdays
 - school_time_blocks
 - grades
 - classes
 - subjects
 - teachers
 - rooms
 - timetable_versions
- يرتبط بـ grade:
 - classes
 - grade_subjects
- يرتبط بـ subject:
 - grade_subjects
 - teacher_subjects
 - subject_time_preferences
 - subject_room_requirements
- يرتبط بـ teacher:
 - teacher_subjects
 - teacher_workload_rules
 - teacher_unavailability
- يرتبط بـ class:
 - class_unavailability
 - class_subject_constraints
 - timetable_entries
- يمتلك timetable_version:
 - timetable_entries
 - generation_runs
 - audit_logs

14) Minimal Required Tables لو أردت نسخة مختصرة فقط

إذا أردت أقل نسخة ممكنة لتشغيل الجدول في البداية، فابدأ بهذه الجداول فقط:

- schools

- academic_years
- terms
- school_timetable_settings
- school_weekdays
- school_time_blocks
- grades
- classes
- subjects
- grade_subjects
- teachers
- teacher_subjects
- teacher_workload_rules
- teacher_unavailability
- class_unavailability
- timetable_versions
- timetable_entries
- timetable_generation_runs
- timetable_generation_issues

The timetable schema must be built around:

- school context
- academic year and term
- configurable weekdays and time blocks
- grade → class → subject structure
- teacher-subject mappings
- workload and availability constraints
- versioned timetable outputs
- diagnostics, issues, and audit logs

The schema must support:

- full generation
- partial regeneration
- AI-based scoring
- conflict detection
- publish/draft/archive versions
- future scalability for rooms, substitutions, and analytics